Data Imputation of Wind Turbine Using Generative Adversarial Nets with Deep Learning Models

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Abstract. Data missing problem is one of the most important issues in the field of wind turbine (WT). The missing data can lead to many problems that negatively affect the safety of power system and cause economic loss. However, under some complicated conditions, the WT data changes according to different environments, which would reduce the efficiency of some traditional data interpolation methods. In order to solve this problem and improve data interpolation accuracy, this paper proposed a WT data imputation method using generative adversarial nets (GAN) with deep learning models. First, conditional GAN is used as the framework to train the generative network. Then convolutional neural network is applied for both the generative model and the discriminative model. Through the zero-sum game between the two models, the imputation model can be well trained. Due to the deep learning models, the trained data imputation model can effectively recover the data with a few parameters of the input data. A case study based on real WT SCADA data was conducted to verify the proposed method. Two more data imputation methods were used to make the comparison. The experiments results showed that the method proposed in this paper is effective.

Keywords: Wind turbine · Data interpolation
Generative adversarial nets (GAN) · Deep learning

1 Introduction

Data missing problem is one of the most significant issues in the industrial field. Due to such problems as sensor faults, communication faults and data storage failure, there have been frequent occurrences of data missing. Complete data plays an important role in WT's operation, maintenance, and fault detection. In the field of wind turbine practice, some data (wind speed, for example) plays such an important role in WT operation and maintenance that the loss of such data may result in failure of power prediction, affect the normal operation of power grid and cause economic losses. Therefore, data imputation is essential in the daily operation in WT industry. Data imputation methods are developed to deal with the problem of data missing.

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In general, the commonly-used method on data imputation is the model-based method, which finds probabilistic distributions of the data and then recovers the missing data from these distributions. In [1], a wavelet-based time hierarchical Bayesian models were build, which can be used to simultaneously model trend. Fabio Orianiae proposed the direct sampling multiple-point statistical technique as a non-parametric missing data simulator for hydrological flow rate time-series [2]. An auto-regressive and moving average (ARMA) method was used as the model to forecast the wind speed data [3].

With the development of artificial intelligence, data-driven methods for data interpolation, neural network (NN) and support vector regression (SVR) for instance, are often used. Zjavka used polynomial neural networks to forecast the wind speed and realize the data imputation [4]. In [5], Petković proposed the fractal interpolation method, which uses back-propagation neural world and extreme learning machine to compensate the wind speed data set. In [6], a hybrid wind speed forecasting model, which combined fast ensemble empirical mode decomposition, sample entropy, phase space reconstruction and back-propagation neural network with two hidden layers, was proposed to enhance the accuracy of wind speed imputation.

However, under some complicated conditions, the accurate of data imputation is limited. In WT field, the data is changing with different environments, in which case, the accuracy of traditional data imputation methods is limited. A simple model cannot deal with non-linear relationships. Therefore, a non-linear data imputation that can work effectively under complicated conditions is needed.

This paper proposed a data imputation method of WT using generative adversarial nets (GAN) [7] based on deep learning models. GAN is a kind of generative model, which has been successfully applied in generating realistic data in many fields [8, 9]. Based on the sound data generation capabilities of GAN, the data imputation of WT can be well conducted. First, the WT data is reasonably prepared for the model training. Then, deep learning networks are used in the architecture of GAN to deal with non-linear relationships. Finally, the trained generative model can be well applied in WT data imputation. A case study based on real WT data was conducted, and the experiment result indicated that the method proposed in this paper is effective.

The rest of this paper is organized as follows. Section 2 is the description of GAN-based WT data interpolation method. Experiments and comparisons are listed in Sect. 3. Section 4 is the conclusion.

2 GAN Based Data Interpolation Method

2.1 Generative Adversarial Nets

GAN [7] is a framework for estimating a generative model via an adversarial process, in which two models are simultaneously trained: a generative model G and a discriminative model D. The generative model G captures the data distribution and the discriminative model D estimates the probability that a sample came from the real data rather than from G. GAN can be trained to generate more realistic data via the zero-sum game. In other words, D and G play the following two-player min-max game with value function $V(G, D)$:

$$\min_G \max_D V(D, G) = \min_G \max_D (E[\log D(x)] + E[\log(1 - D(G(z)))]) \tag{1}$$

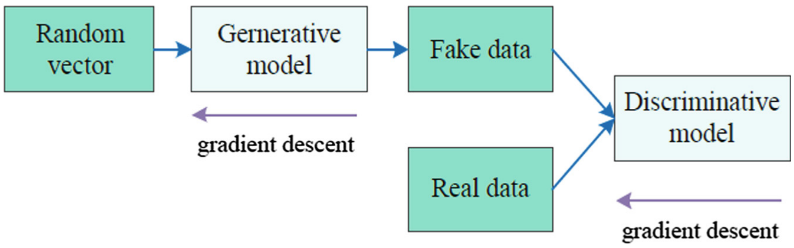

Fig. 1. Schematic diagram of the original GAN.

Figure 1 is the schematic diagram of the original GAN. First, a random vector is transformed into the fake data by the generative model. Then the real data is involved, and combined as the input of the discriminative model. Third, the discriminative model distinguishes the real data from the fake data. The parameters of the generative model and the discriminative model are updated by the gradient descent method in every epoch during the model training. Finally, after the model training, the generative model can be obtained.

2.2 Interpolation Method

However, in the case of data imputation, in order to impute the missing data correctly, the known data is used to recover the missing data, which is a kind of supervised learning. The original GAN is unsupervised learning, so it cannot process the data in a supervised way. Therefore, a supervised GAN is needed.

Conditional GAN is developed as a supervised learning, so that the generated data can be obtained in a supervised way. Conditional GAN is used as the framework of WT data imputation. Moreover, in order to get a better data imputation model, deep learning networks are used in the framework of GAN. In this paper convolutional neural network is chosen.

For the training data, the WT data should be prepared in a proper way to train the deep learning network. In this paper, other known WTs data is used to impute the missing data of a certain WT. The data organization flowchart is shown in Fig. 2.

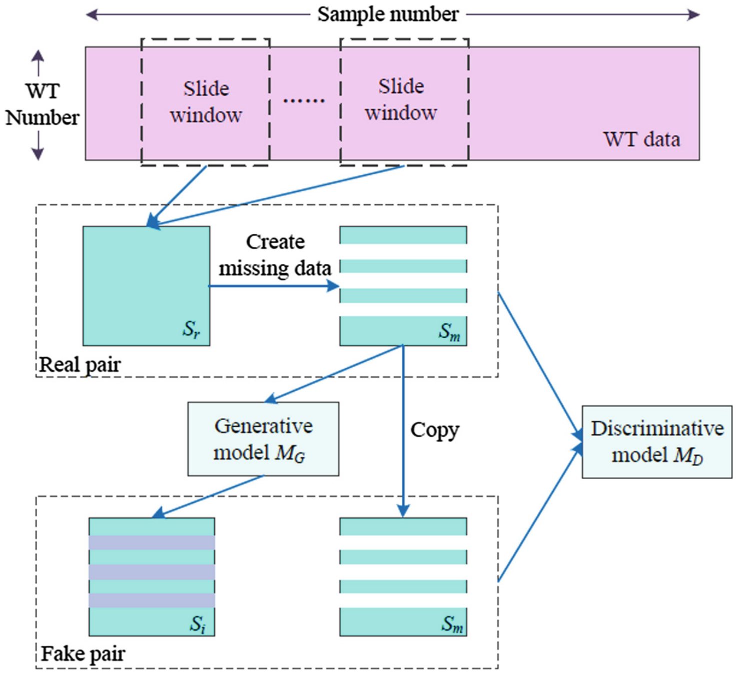

Fig. 2. Flowchart of WT data imputation.

Data Organization. The WT data is collected and organized as the training samples of the data imputation model.

First, in order to provide the training data with more useful information, the collected data from each WT are put together. Suppose d_{ij} is the collected data from i^{th} WT, where i is the WT number and j is the sample number. A training data can be organized as a matrix:

$$D = \begin{bmatrix} d_{11} & d_{12} & \cdots & d_{1n} \\ d_{21} & d_{22} & \cdots & \\ \vdots & \vdots & \vdots & \vdots \\ d_{m1} & d_{m2} & \cdots & d_{mn} \end{bmatrix} \tag{2}$$

where m is the number of WTs and n is the sample numbers of the data.

Second, a slide window δ is constructed to get every training sample. The height of δ is equal to m and the width of δ is equal to n. Then the data samples S_r can be obtained by moving δ from the beginning to the end of the WT data.

Third, the data S_r in some specified rows (the data of specified WTs) are eliminated. The eliminated data S_m are considered as the missing data. So the real data pairs (S_r, S_m) of the training data are obtained.

$$S_m = S_r \times R_i = \begin{bmatrix} d_{11} & d_{12} & d_{13} & \cdots & d_{1n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ d_{1i} & d_{2i} & d_{3i} & \cdots & d_{ni} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ d_{m1} & d_{m2} & d3 & \cdots & d_{mn} \end{bmatrix} \times R_i = \begin{bmatrix} d_{11} & d_{12} & d_{13} & \cdots & d_{1n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ d_{m1} & d_{m2} & d3 & \cdots & d_{mn} \end{bmatrix} \tag{3}$$

where R_i is an identity matrix of n in which i^{th} row is all zeros. In practice, multiple rows of S_r may be deleted and accordingly multiple R_i may be used.

Model Training. GAN is used as the framework and convolutional neural networks are selected as the generative and discriminative models, by which the data imputation model can be trained.

First, the missing data S_m is processed by the generative model M_G of GAN, by which the imputed data S_i can be obtained. So the fake data pairs *(S_i, S_m)* of the training data are obtained.

Then, the real data pairs *(S_r, S_m)* and the fake data pairs *(S_i, S_m)* are used as the input of the discriminative model M_D of GAN. M_D tries to identify the real data pair from two data pairs. In order to make the imputed data close to the ground truth, a traditional loss is added in *L(M_G)* and L1 distance is used to measure the distance of S_r and S_i. The loss function of M_D is:

$$L(M_D) = \mathrm{E}[\log M_D(S_r, S_m)] + \mathrm{E}[\log(1 - M_D(S_i, S_m))] \tag{4}$$

and the loss function of M_G is:

$$L(M_G) = \mathrm{E}[\log(1 - M_D(S_i, S_m))] + \lambda ||S_r - S_i||_1 \tag{5}$$

Third, parameters of M_G are fixed, and parameters of M_D are updated according to the result. Accordingly, after the updating parameters of M_D, parameters of M_D are fixed, and the parameters of M_G are updated. The parameter updating of M_D is:

$$\theta^{(n+1)}(M_D) = \theta^{(n)}(M_D) - \eta \nabla_{M_D} L(M_D) \tag{6}$$

and parameter updating of M_G is:

$$\theta^{(n+1)}(M_G) = \theta^{(n)}(M_G) - \eta \nabla_{M_G} L(M_G) \tag{7}$$

where $\theta^{(n+1)}$ is the parameters of $(n+1)^{th}$ iteration of the training and $\theta^{(n+1)}$ is the parameters of n^{th} iteration. ∇ is the gradient decent algorithm and η is the learning rate.

Through the zero-sum game of the two models, the data imputation model can be obtained.

Evaluation Indicators. In order to make a clear comparison, Mean Absolute Error (MAE), Mean Squared Error (MSE) and Normalized Root Mean Squared Error (NRMSE) are selected as the evaluation indicators of the effectiveness of interpolation.

$$MAE = \frac{1}{n}\sum_{t=1}^{n} |y_t - \bar{y}_t|^2 \tag{8}$$

$$MSE = \frac{1}{n}\sqrt{\sum_{t=1}^{n} (y_t - \bar{y}_t)^2} \tag{9}$$

$$NRMSE = \frac{\sqrt{\sum_{t=1}^{n} (y_t - \bar{y}_t)^2}}{\sum_{t=1}^{n} \bar{y}_t^2} \tag{10}$$

where y_t is the imputed data and $\bar{y}_t$ is the real data. n is the length of the missing data.

3 Case Study

In this section, the proposed method is verified by experiments with real WT data. The method proposed by this paper is compared with ARMA method and BPNN method.

3.1 Experiment Settings

A wind farm in northern China is selected for this study. This wind farm kept a good record of the supervisory control and data acquisition (SCADA) data in recent five years. The main components of the WT and its collected variables of the SCADA data are shown in Fig. 3. We collected the SCADA data from 34 WTs of the wind farm, and the wind speed is chosen as the interpolation target in this study.

In this study, the data of some normal WTs were used to impute the data of the data-missing WT (for example, suppose the wind speed data of WT06 is missing, and the wind speed data of other WTs is used to impute the data of WT 06). In order to test the interpolation methods, real missing data was not used in experiments. Instead, the complete wind speed data was used. The wind speed data of some certain WTs were eliminated artificially. Then the interpolation methods were used to recover the eliminated data.

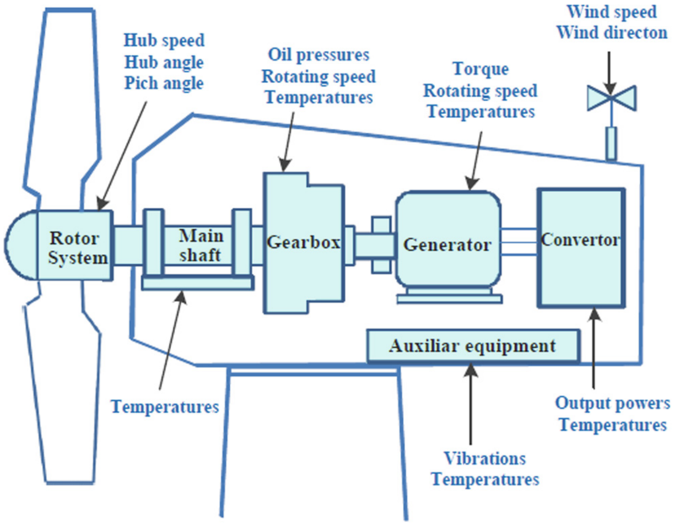

Fig. 3. Main components of WT and main variables of the collected SCADA data.

In this experiment, another two methods were used to make a comparison with the proposed method. The ARMA method was used as a time series method of the regression algorithm, and the back propagation neural network (BPNN) method was chosen as a neuron network method of the machine learning algorithm.

In order to test the data imputation effect of the method, an extreme experiment setting was used. The wind speed data of WT 03, WT 07, WT 011, WT 15, WT 19, WT 23, and WT 27 were kept and wind speed data of the other WTs were all eliminated. The three methods (ARMA method, BPNN method and the proposed method) were conducted to impute the missing data.

3.2 Comparative Experiments

In these experiments, the complete data and the eliminated data were combined as the training data. Another 300 eliminated data was used as the testing data. The data imputation results of WT 12 and WT 18 are shown in Figs. 4 and 5. The three subfigures respectively show the results of the ARMA method, BPNN method and the proposed method.

As can be seen, due to the influence of the environment, the wind speed changes dramatically. Moreover, there are fluctuations in the data. It increases the difficulty of data interpolation. From the data imputation result, the trend of the recovered data is basically correct with the ARMA method and BPNN method. However, in detail, due to data fluctuations and data changing in different environments, the data imputation

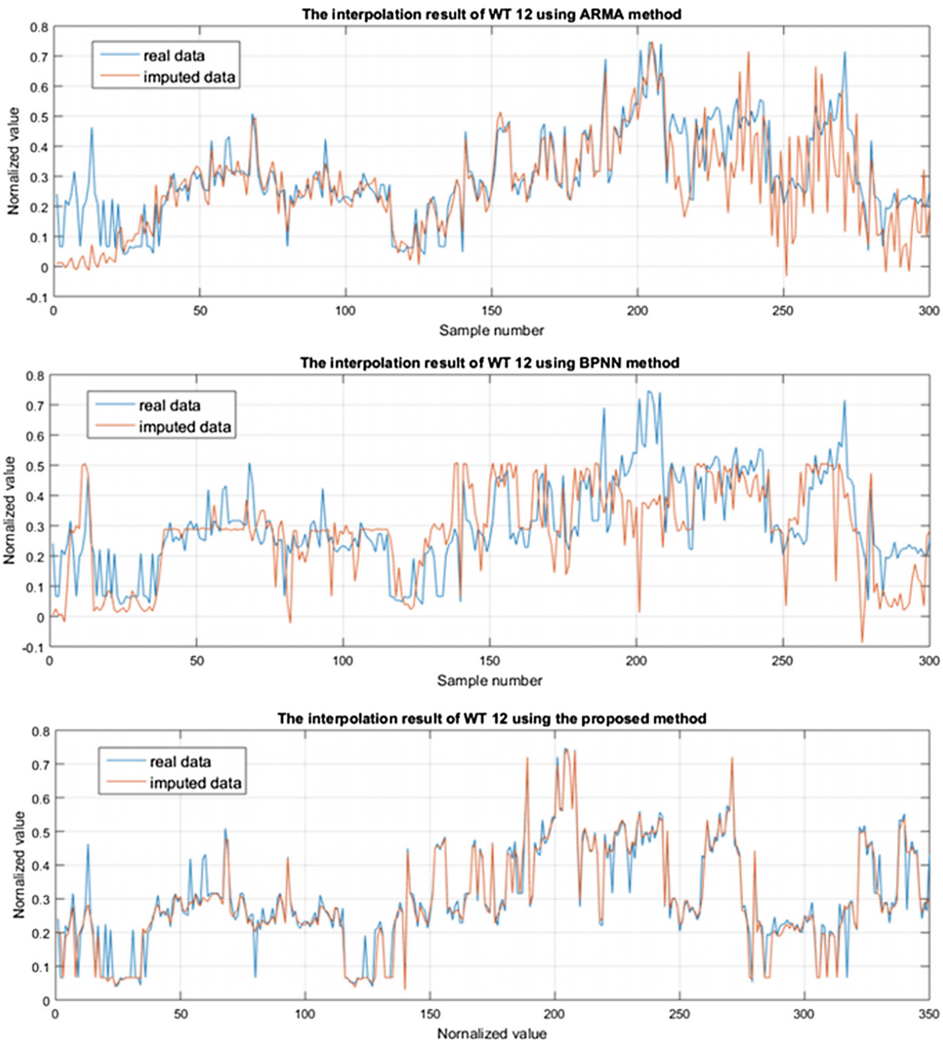

Fig. 4. Data imputation results of WT 12 using ARMA method, BPNN method and the proposed method.

effect is not so good. By contrast, given the same condition, the effect of the proposed method is more accurate in both the data trend and the detail. The result indicates that the proposed method can effectively impute WT wind speed data under such conditons.

The detailed data imputation results of WT 12, WT 18 and WT 24 are listed in Table 1. In can be seen that the proposed method has a good performance in all the three indicators of the data imputation.

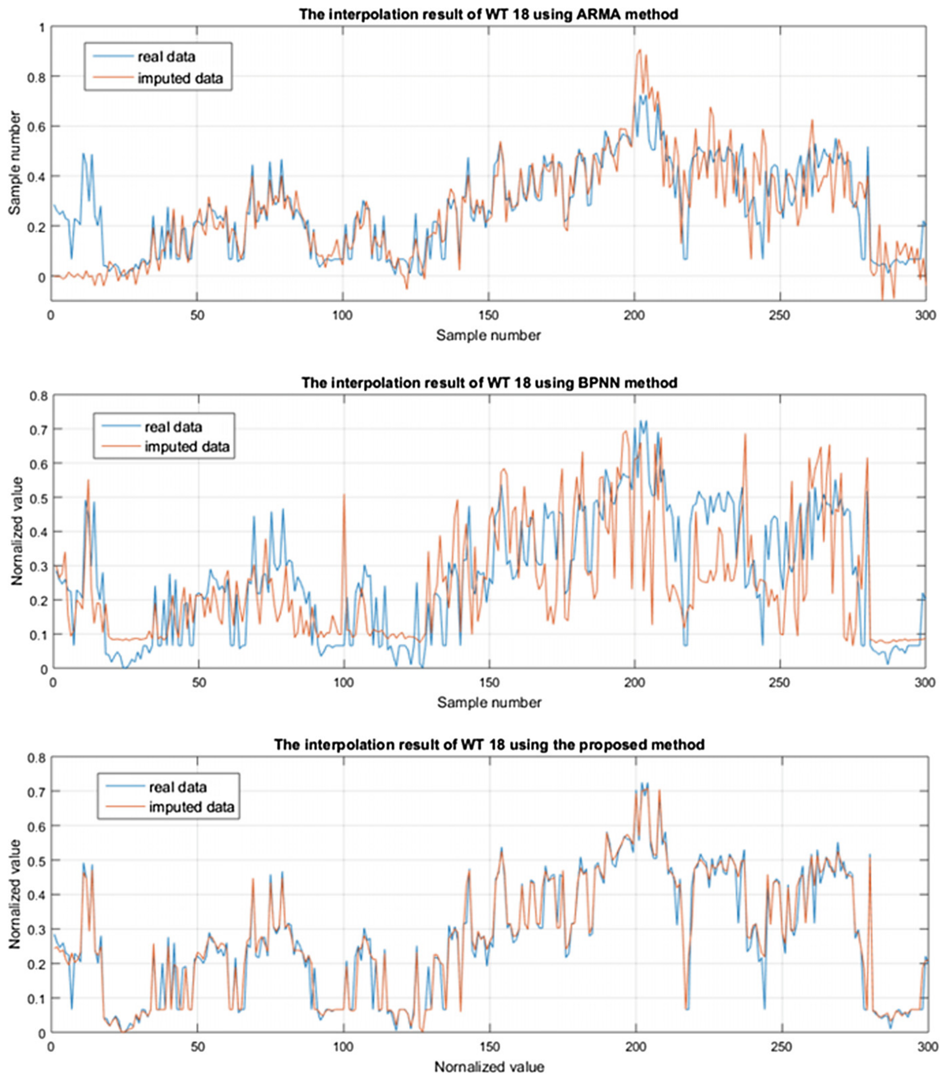

Fig. 5. Data imputation results of WT 18 using ARMA method, BPNN method and the proposed method.

Table 1. Data imputation results.

	ARMA	BPNN	The proposed method
MAE (WT 12)	0.0809	0.1035	0.0180
MSE (WT 12)	0.0084	0.0080	0.0023
NRMSE (WT 12)	0.0246	0.0263	0.0076
MAE (WT 18)	0.0801	0.1452	0.0177
MSE (WT 18)	0.0081	0.0106	0.0023
NRMSE (WT 18)	0.0253	0.0390	0.0084
MAE (WT 24)	0.0927	0.1002	0.0169
MSE (WT 24)	0.0089	0.0079	0.0022
NRMSE (WT 24)	0.0287	0.0277	0.0077

4 Conclusion

Data missing problem is one of the most significant study topics in the field of WT. The collected data is often lost due to connection faults, storage problems, software errors and other reasons. Data missing can lead to many problems, which not only threatens the security of the power system but also affect the economic benefits. In the field of WT, the collected data is changing in different environments, so it is difficult to impute the missing data with high accuracy. In order to solve this problem, this paper proposed a WT data imputation method based on GAN with deep learning models. GAN is used as the architecture, and the zero-sum game between the generative model and discriminative model can make the generator generating data more realistic. Then convolutional neural network is used for both the generative model and the discriminative model. Finally, a case study based on real WT data was conducted. Through the comparison using ARMA method and BPNN method, the effectiveness of the proposed method is proved.

Further research of this study will focus on the training data of the GAN. A good preparation of the training data can ensure the accuracy of the data imputation. More research about this topic will continue to be studied.

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